Final Project EDT

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1 Phase 1: Problem Definition and Data Acquisition

1.1 Event Definition and Study Area

This project revisits the case study analyzed in the Midterm Project: the collapse of the Kakhovka Dam on June 6, 2023, located on the Dnieper River in Kherson Oblast, southern Ukraine. The event caused catastrophic downstream flooding and the rapid drainage of the Kakhovka Reservoir, making it one of the most significant hydrological disasters in recent European history.

The Region of Interest (ROI) is centered on the area surrounding Nova Kakhovka, where the effects of reservoir drainage were most clearly observable in satellite imagery. This ROI was selected based on image availability and the visibility of hydrological changes between the pre- and post-event dates.

1.2 Satellite Imagery Metadata

Two Sentinel-2 Level-2A images were acquired from the Copernicus Open Access Hub. Table 1 summarizes their main technical characteristics.

Table 1: Technical characteristics of the satellite imagery used.

Date	Satellite	Level	Resolution	Cloud Cover	Role
June 18, 2023	Sentinel-2 A/B	Level-2A	10 m	$<$30%	Post-event

Level-2A products provide Bottom-of-Atmosphere (BOA) reflectance, eliminating the need for additional atmospheric correction. The bands used for this project span from the visible to the shortwave infrared spectrum, as detailed in Table 2.

Table 2: Sentinel-2 spectral bands used for dataset construction.

Band	Name	Wavelength (μm)
B2	Blue	0.490
B3	Green	0.560
B4	Red	0.665
B5	Red Edge 1	0.705
B6	Red Edge 2	0.740
B7	Red Edge 3	0.783
B8	Near-Infrared (NIR)	0.842
B8A	Narrow NIR	0.865
B11	SWIR 1	1.610
B12	SWIR 2	2.190

1.3 Dataset Construction

To prepare the training dataset for the machine learning phase, pixel values were extracted from the Sentinel-2 band stack at the locations defined by the training polygons digitized in QGIS. Each polygon corresponds to one of four land cover classes: Water, Vegetation, Bare Soil, and Built-up.

The extraction was performed using a Python script based on the `rasterio` and `geopandas` libraries. For each pixel falling within a training polygon, the script records its geographic coordinates (Latitude, Longitude), the reflectance values for all ten spectral bands, and the corresponding class label. The output was exported as a Tab-Separated Values (TSV) file, as required for the subsequent training phase.

Table 3 illustrates the structure of the resulting dataset.

Table 3: Structure of the exported TSV training dataset.

Column	Type	Description
Latitude	Float	Geographic latitude of the pixel centroid
Longitude	Float	Geographic longitude of the pixel centroid
B2	Float	Blue band reflectance
B3	Float	Green band reflectance
B4	Float	Red band reflectance
B5	Float	Red Edge 1 reflectance
B6	Float	Red Edge 2 reflectance
B7	Float	Red Edge 3 reflectance
B8	Float	NIR band reflectance
B8A	Float	Narrow NIR reflectance
B11	Float	SWIR 1 reflectance
B12	Float	SWIR 2 reflectance
label	String	Land cover class (Water, Vegetation, Bare Soil, Built-up)

2 Phase 2: Pre-processing and Training

2.1 Pre-procesing

2.2 Models

2.2.1 Decision Trees (DT)

2.2.2 Support Vector Machine (SVM)

A Support Vector Machine (SVM) classifier was implemented to perform supervised land cover classification using the spectral information extracted from the Sentinel-2 imagery. SVM is a widely used machine learning algorithm in remote sensing applications due to its strong performance in high-dimensional datasets and its ability to separate classes using optimal decision boundaries.

The training dataset, evenly distributed among the four land cover classes: Built-up, Soil, Vegetation, and Water. Each sample corresponds to a single pixel and contains the reflectance values from the ten spectral bands used in this project.

Before training, the dataset was divided into training and testing subsets using a stratified split in order to preserve the class distribution. Approximately 70% of the samples were used for training and 30% for evaluation.

The SVM model achieved an Overall Accuracy of 99.75%, demonstrating excellent classification performance across all land cover categories. Precision, recall, and F1-score values were close to 1.00 for every class, indicating that the spectral signatures extracted from the Sentinel-2 bands provide strong class separability.

Table 4 summarizes the evaluation metrics obtained during testing.

Table 4: Classification report for the SVM model.

Class	**Precision**	**Recall**	**F1-score**	**Support**
Built-up	1.00	1.00	1.00	608
Soil	1.00	0.99	1.00	608
Vegetation	0.99	1.00	1.00	609
Water	1.00	1.00	1.00	608

The confusion matrix shown in Figure 1 confirms the high classification accuracy of the model. Most samples were correctly classified, with only a very small number of misclassifications between spectrally similar classes.

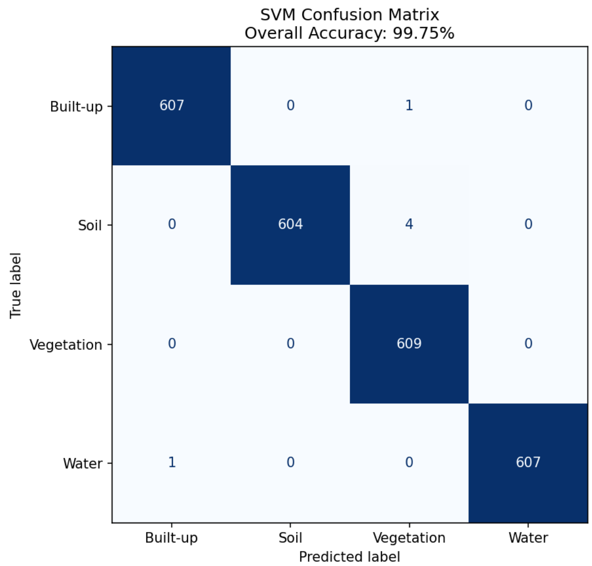

Figure 1: Confusion matrix obtained from the SVM classifier.

These results indicate that the SVM model is highly effective for multispectral land cover classification using Sentinel-2 imagery and provides a strong baseline for comparison with the other machine learning approaches evaluated in this project.

2.2.3 Artificial Neural Networks (ANN)

An Artificial Neural Network (ANN) model was implemented to evaluate the performance of deep learning techniques for multispectral land cover classification. Neural networks are capable of learning complex nonlinear relationships between spectral features and target classes, making them highly suitable for remote sensing applications involving high-dimensional data.

The ANN was trained using the same dataset employed for the SVM classifier, containing 8108 labeled pixel samples equally distributed among the classes Built-up, Soil, Vegetation, and Water. Each sample contains the reflectance values corresponding to the ten Sentinel-2 spectral bands selected for this study.

Prior to training, the dataset was normalized and divided into training and testing subsets using a stratified split. Approximately 70% of the data was used for training, while the remaining 30% was reserved for model evaluation.

The neural network architecture consisted of multiple fully connected dense layers with ReLU activation functions and Dropout regularization layers to reduce overfitting. The final output layer used a softmax activation function to perform multiclass classification across the four land cover categories.

The model was trained for 50 epochs and achieved a final Overall Accuracy of 99.59% on the testing dataset. The training history demonstrated stable convergence and high validation accuracy throughout the training process.

Table 5 summarizes the classification metrics obtained for the ANN model.

Table 5: Classification report for the ANN model.

Class	**Precision**	**Recall**	**F1-score**	**Support**
Built-up	1.00	1.00	1.00	608
Soil	0.99	1.00	0.99	608
Vegetation	1.00	1.00	1.00	609
Water	1.00	0.99	1.00	608

Figure 2 presents the confusion matrix generated from the ANN predictions. The matrix indicates that the vast majority of samples were correctly classified, with only minimal confusion between a few spectrally similar classes.

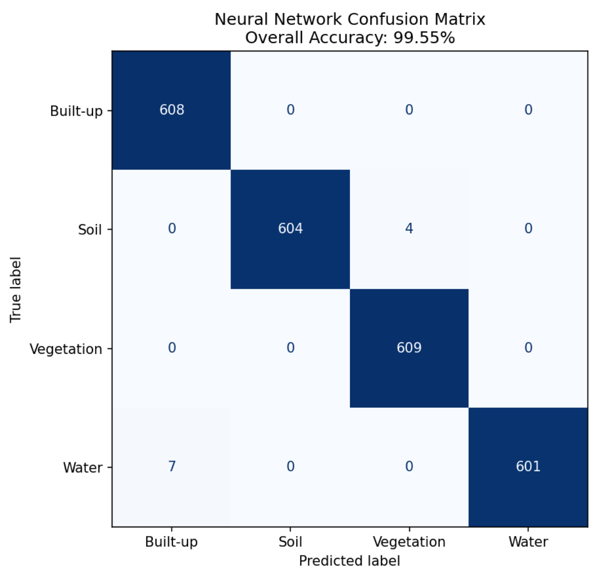

Figure 2: Confusion matrix obtained from the ANN classifier.

Figure 3 shows the training history of the neural network, including the evolution of training and validation accuracy during the learning process. The results demonstrate rapid convergence and stable generalization performance.

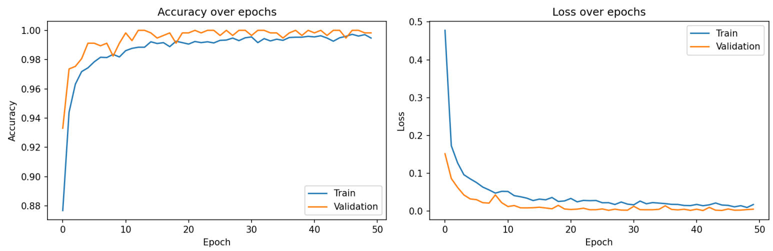

Figure 3: Training and validation accuracy history for the ANN model.

The ANN model achieved performance comparable to the SVM classifier, confirming that deep learning approaches can effectively model the spectral behavior of different land cover types using Sentinel-2 multispectral imagery.

2.2.4 K-Nearest Neighbor (KNN)

2.2.5 Naive Bayes (NB)

3 Phase 3: Synthesis and Prediction